

AI Elite Program — Customer Presentation Document

Autonomous IBM Power Porting Triage Agent

Initiative: AI Elite Program Prototype Submission	Target Architecture: IBM Power (ppc64le)
Persona: Technical Pre-Sales & Migration Specialists	Runtime Platforms: RHEL 8/9, OpenShift (OCP), Ubuntu, SLES

1. Customer & Target Persona

Customer Profile: Global enterprises running high-throughput transactional, time-series, and data/AI workloads (banking, telecom, retail, and healthcare) evaluating migration from commodity x86 infrastructure to **IBM Power (ppc64le)** on Red Hat Enterprise Linux and OpenShift.

Target Personas:

- **IBM Technical Pre-Sales Engineers & Solution Architects:** Frontline technical sellers who must rapidly qualify migration feasibility, size porting timelines, and build customer confidence during competitive evaluations.
- **Enterprise Architecture Decision-Makers (CIO / VP Infrastructure):** Stakeholders requiring risk transparency, timeline estimates, and platform ROI before signing migration contracts.
- **IBM Power Porting Lab & Ecosystem Engineers:** Specialized engineering squads responsible for compiling, tuning, and verifying unported packages.

2. Business Problem & Estimation Gap

While IBM Power offers unmatched hardware throughput (high memory bandwidth, SMT8 core threading, and hardware-accelerated Matrix Math Accelerators), **pre-sales qualification creates severe friction in the sales pipeline:**

- **Unstandardized & Heterogeneous Inputs:** Customers provide dependency specifications in fragmented formats: CycloneDX/SPDX SBOM JSONs, multi-stage Dockerfiles, Python requirements.txt, or unstructured meeting notes.
- **Fragmented Ecosystem Registries:** Determining package availability requires tedious manual searches across RHEL BaseOS/AppStream, EPEL 9, Quay.io, Docker Hub, PyPI, and community channels (Conda-forge, IBM Open-CE).
- **The 'Iceberg' Estimation Gap:** When a component lacks a ready binary, pre-sales teams cannot easily see beneath the surface. Uncovered build-time dependencies (e.g. jemalloc needing 64KB page size tuning) or unported x86 SIMD intrinsics (AVX2/AVX-512) surface late, resulting in costly project overruns.

3. Quantified Business Impact

Operational Dimension	Current Manual Process	AI Agent Prototype Impact
Qualification Turnaround Time	5 to 10 Business Days of email exchanges & manual catalog lookups	< 30 Seconds automated end-to-end qualification
Porting Lab Bandwidth Waste	~40% of senior engineer time spent on routine initial triage	Zero wasted hours on routine triage; lab only receives scoped specs
Deal Velocity & Friction	Prolonged sales cycles; customer hesitation due to unknown porting risks	Immediate sales confidence with instant traffic-light readiness scoring
Effort Sizing Accuracy	Subjective guesswork prone to 2x–5x variance from hidden dependencies	Standardized 3-Pillar Sizing Formula scoping hardware friction & build trees
Executive Proposal Deliverables	Days spent assembling custom slide decks and proposal memos	1-Click Executive Memo & CSV Backlog generated autonomously

4. Proposed Opportunity: Autonomous AI Agent Architecture

We developed a production-ready, containerized AI agent combining deterministic ecosystem probing with generative sales synthesis:

- **Universal Manifest Ingestion Engine:** Automatically parses CycloneDX/SPDX SBOMs, Dockerfiles, Python requirements, and raw text into normalized component representations across Containers, C/C++, PyPI, RPM, and Java.
- **Multi-Source Deterministic Prober:** Executes real-time live queries against Quay.io, Docker Hub v2, RHEL/EPEL 9 repos, and PyPI to verify multi-arch ppc64le support, returning canonical, clickable verification URLs.
- **Transitive Dependency Iceberg Scoper:** Recursively analyzes unported packages to uncover hidden build-time requirements (BuildRequires, compiler toolchains, submodules).
- **Hardware Architecture Sensitivity Audit:** Detects x86 SIMD/AVX2 intrinsics (prescribing SIMDe to Power VSX translations), 64KB page size sensitivities (jemalloc tuning), and inline assembly.
- **Standardized 3-Pillar Sizing Formula:** Computes Person-Days as Base Compilation + Transitive Sub-deps + Architecture Friction, incorporating a 1-week deployment validation baseline.
- **Generative Executive Proposal Engine:** Uses Google Gemini (with resilient multi-model fallback across gemini-3.5-flash and gemini-3.5-flash-lite) to draft client-ready strategic value proposals highlighting Power hardware advantages.

5. Demonstrated Workload Scenarios & Validation

Scenario / Workload	Stack Components	Agent Triage Findings	Score & Sizing
Enterprise Cloud Microservices	Nginx, Redis, PostgreSQL, Strimzi Kafka, Node.js	All components have verified native ppc64le container images in Docker Hub and Quay.io.	100% GO 5 Person-Days
AI / ML Vector Pipeline	PyTorch, NumPy, Pandas, Scipy, Custom C++ DSP	PyTorch routed to IBM Open-CE with Power MMA acceleration; custom DSP kernel scoped with SIMDe AVX2->VSX fix.	95% GO 5 – 8 Person-Days
High-Throughput Storage Engine	RocksDB 8.6, simdjson 3.6, Redis, zlib	Deep dependency iceberg scoped: 6 build deps uncovered, including unported 64KB-page tuned jemalloc and AVX-512 kernel.	30% CAUTION 8 – 12 Person-Days
Legacy Financial Analytics	Intel MKL, Python 3.9, Proprietary Risk Calc	Proprietary closed-source x86 blocker flagged; automated recommendation to substitute with IBM ESSL or OpenBLAS.	HIGH RISK Arch substitution

6. Strategic Value & Program Alignment

This prototype directly addresses the core mission of the **AI Elite Program** by applying advanced agentic AI to eliminate real-world enterprise sales friction:

1. **Accelerates Infrastructure Revenue:** Eliminates the 10-day pre-sales bottleneck, arming client teams with authoritative migration roadmaps within seconds of first contact.
2. **Optimizes Engineering Allocation:** Prevents the IBM Power Porting Lab from wasting bandwidth on off-the-shelf components, focusing scarce specialist time exclusively on scoped unported libraries.
3. **Demonstrates Hybrid AI Leadership:** Seamlessly combines deterministic knowledge-graph probing, container registry APIs, and multi-model LLM reasoning to solve mission-critical systems challenges.

Submission Note: *Prototype is fully implemented, containerized via Podman/Docker Compose, tested with 100% passing test suites, and available for live demonstration.*